



LST Administration of Members and Authors

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1 Introduction

The LST Administration of Members and Authors (LAMA) system is a web-based platform developed using Django for the LST Collaboration. It serves as a tool for managing the roles, duties, memberships, and authorship statuses of collaboration members.

This document offers a comprehensive overview of LAMA, detailing its core functionalities, system architecture, and operational guidelines.

The LAMA system provides:

- A centralized database for managing member information, including names, email addresses, institutional affiliations.
- Tracking of:
 - Duties assigned to members,
 - Common Fund (CF) contribution periods,
 - Membership durations,

which together determine authorship eligibility according to predefined rules.

- Automatic computation of authorship periods based on duties and CF contributions.
- Automatic creation and maintenance of `Authordetail` records to ensure consistency in authorship tracking.
- Export functionality for generating an up-to-date author list.

Additionally, LAMA includes a statistics module with:

- Visual dashboards displaying key summary metrics.
- Filtering options by country, group, and institute.
- Historical data visualizations using interactive histograms and time-based trends.

NOTE: Only users with elevated permissions can modify or add new data. If you identify an error, please send a change request to lst-telescope-manager@cta-observatory.org

2 Login

The LST Administration of Members and Authors system utilizes LDAP authentication to manage user access securely. Users can log in to the system using their CTAO credentials.

Upon opening LAMA webpage, the user is welcomed by a general page visible in Fig. 2.1. Click on the button "Log In To Get Started!" to be redirected to the login page.

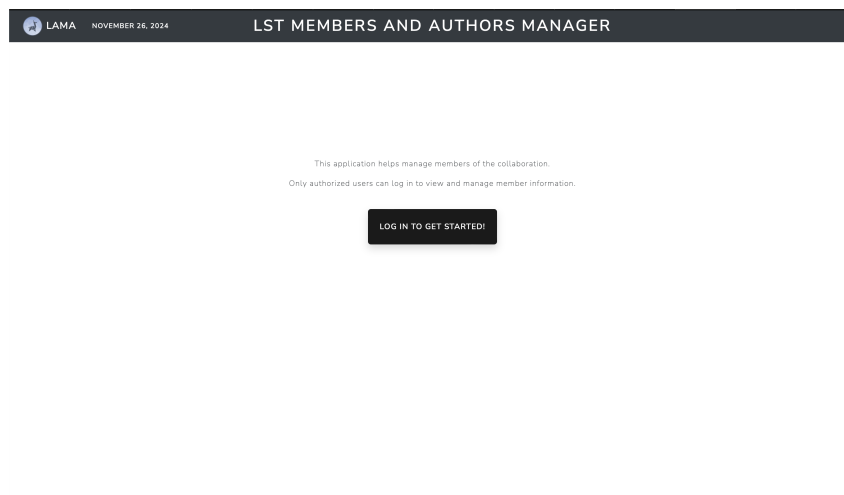


Figure 2.1 – Start view.

2.1 Login Process

To log in, users must follow these steps:

1. Navigate to the login page of the LAMA system (Fig. 2.2).
2. Enter your username in the format `name.surname`. This should match the CTAO credentials username.
3. Enter your CTAO credentials password in the provided password field.
4. Click on the "Login" button to authenticate.

Once successfully authenticated, users will be redirected to the home page.

2.2 LDAP Authentication

The system communicates with the CTAO LDAP server to verify user credentials. When a user attempts to log in, their username and password are sent to the LDAP server for validation. If the credentials are correct, the user is granted access to the LAMA system.

The authentication is handled through the following LDAP structure:

- Users must enter their `name.surname` as the username.

Figure 2.2 – Login page.

- The password field must match the user's CTAO credentials password.
- The LAMA system uses this information to authenticate against the organization's central LDAP server.

2.3 Permissions and Roles

Permissions are linked to specific LDAP groups, and LAMA assigns user permissions based on their group affiliation. Permissions control the level of access a user has within the system, such as viewing, editing, or managing members and authors. In LAMA, two roles are defined, with their corresponding permissions as follows:

- **Admin:** Full access to all features of the system, including managing users, members, authors, and statistics. Admin users can also assign or revoke permissions for other users.
- **Sapo:** Semi-administrative access, with the ability to edit and create institutes, manage author details for publications. Sapo users cannot manage users or system settings.
- **Viewer:** Can view data and statistics, but does not have permission to make any changes to the system. Viewers have read-only access to the system's dashboards and statistics.

2.3.1 LDAP Group Mapping

Each role within the LAMA system is mapped to an LDAP group, which determines the permissions granted to the user. When a user logs in, the system checks their LDAP group membership and grants the corresponding permissions, assigning the user to the correct role.

- **lst-membership-admin:** Members of this group are granted Admin permissions.
- **lst-sapo:** Members of this group are granted Sapo permissions.

- **lst-members:** Users in this group are granted Viewer permissions.

The system automatically checks the user's LDAP group upon login and assigns the corresponding permissions. If a user belongs to multiple groups, the highest level of permission is granted.

2.4 Password Management

Since LAMA relies on LDAP for authentication, password management is handled by the CTAO system. Users should follow the organization's guidelines for password resets and changes.

2.5 Security Considerations

To ensure the security of user data:

- The system does not store passwords; it only verifies them against the LDAP server.
- Sessions are automatically timed out after a 30-minute period of inactivity.

2.6 Logging Out

To log out of the system, users can click the "Logout" button in the top right corner of the page. Logging out will terminate the session and require re-authentication to access the system again.

3 Usage Guide

3.1 Navbar

Throughout the application, the navigation bar (Fig. 3.1) is displayed at the top of the interface, providing key tabs to navigate the site.



Figure 3.1 – Navigation bar of the app in case of Admin permission, with the added button "Admin Panel" compared to the standard permission level.

The navigation bar includes the following elements:

- **Admin Panel:** A button that brings the user to the Django admin site. This is only available to users with Admin permissions.
- **Username:** Displays the logged-in user's name for easy identification.
- **Logout:** Allows the user to log out of the application and end their session.

The navigation menu, located at the center of the navigation bar, includes the following tabs:

- **Home:** Returns the user to the main page (the Home Page).
- **Members:** Provides access to the Members List page, where users can view and manage member information.
- **Authors:** Provides access to the Authors List page, where users can view and edit author details for publications.
- **Institutes:** Opens the Institutes list page, allowing users to view, edit, and create institutes.
- **Duties:** Redirects users to the Duty page, which provides information about each member's duty in the member list and duty information in the duty list.
- **Statistics:** Redirects users to the Statistics page, which provides detailed visual dashboards and trends.

The currently active tab is highlighted for easy navigation. Clicking on any of the other tabs will take the user to the corresponding page.

3.2 Home Page

Once a user has logged into LAMA, they will be directed to the **Home Page**, shown in Fig. 3.2. The Home Page serves as the central hub, providing users with a quick overview of the collaboration's status.

This page presents the following summary metrics, allowing users to quickly grasp key information about the collaboration:

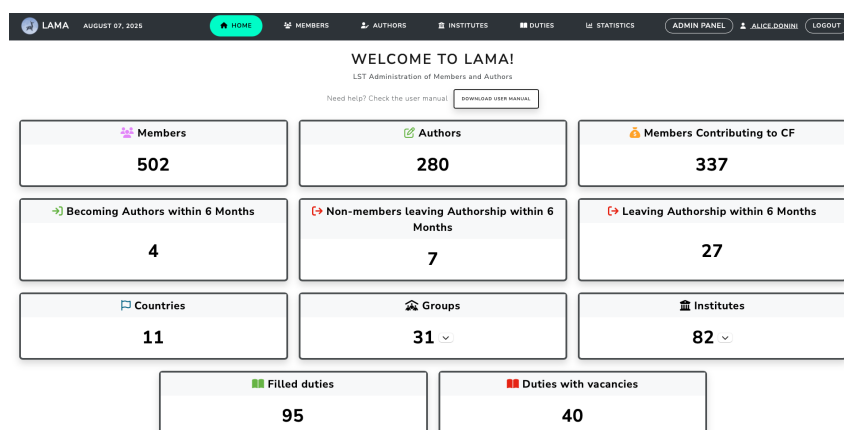


Figure 3.2 – Home view.

- Total number of active members and authors.
- Number of members contributing to the Common Fund (CF).
- Number of members who will become authors within the next 6 months.
- Number of former members who are still authors (with a max duration of 6 months).
- Number of members who will leave authorship within the next 6 months.
- Number of countries, groups, and "official" institutes involved in the project.
- Number of duties that are fully assigned (i.e., all available member slots are filled).
- Number of duties that are partially assigned (i.e., the number of assigned members is below the maximum allowed).

These key statistics are presented in summary cards for an at-a-glance overview of the most relevant data.

3.3 Members Page

The **Members Page**, displayed in Fig. 3.3, is where users can view and manage the list of members and their associated details. It is one of the main functionalities of the LAMA system, and it provides a detailed view of all members, their roles, and key information about their status within the collaboration.

The table on the Members Page includes several filters on the top left (Fig. 3.4) that help users refine the displayed data. These filters allow users to search for members based on specific criteria. There is also a search bar that enables searching for any written pattern. By default, the table shows only active members, excluding former members. To include past members, users can use the toggle switch . When active, the toggle switch turns green .

Filters are designed to be simple, offering drop-down selection options for the following criteria:

- **Country Filter:** Users can select a country from a drop-down list. The page will display only members associated with that country.

Admin view of the Members Page. The table displays 10 members with the following data:

#	FIRST NAME	LAST NAME	INSTITUTE	GROUP	COUNTRY	MEMBERSHIP START	MEMBERSHIP END	CF	AUTHOR	AUTHORSHIP START	AUTHORSHIP END	ACTIONS
1	Abelardo	Moralejo	IFAE-BIST	IFAE	Spain	2018-01-01	-	Yes	Yes	2018-07-01	-	[Edit] [Delete] [Details]
2	Abhishek	Abhishek	INFN Siena	INFN	Italy	2023-03-01	-	Yes	Yes	2023-09-01	-	[Edit] [Delete] [Details]
3	Abhadeep	Roy	U Hiroshima	JAPAN	Japan	2024-02-01	-	Yes	Yes	2024-08-01	-	[Edit] [Delete] [Details]
4	Alon	Viana	IFSC-USP	IFSC-USP	Brazil	2024-12-16	-	Yes	Yes	2025-06-16	-	[Edit] [Delete] [Details]
5	Akiko	Takagi	U Tokyo, ICRR	JAPAN	Japan	2023-09-15	-	No	No	-	-	[Edit] [Delete] [Details]
6	Akira	Okumura	U Nagoya, ISEE	JAPAN	Japan	2018-01-01	-	Yes	Yes	2018-07-01	-	[Edit] [Delete] [Details]
7	Alan	Sunny	IAPS Roma	INAF	Italy	2024-01-01	-	Yes	Yes	2024-07-01	2026-12-31	[Edit] [Delete] [Details]
8	Alba	Moreno Ramos	UCM, Electronica	UCM-ELEC	Spain	2024-04-01	-	Yes	No	-	-	[Edit] [Delete] [Details]
9	Albert	Kong	U Tokyo, ICRR	JAPAN	Japan	2018-01-01	-	Yes	No	-	-	[Edit] [Delete] [Details]
10	Alberto	Rosales de Leon	U Lodz	LOOZ	Poland	2025-02-10	-	No	No	-	-	[Edit] [Delete] [Details]

(a) Admin view.

User view of the Members Page. The table displays 10 members with the following data:

#	FIRST NAME	LAST NAME	EMAIL	INSTITUTE	GROUP	COUNTRY	MEMBERSHIP START	MEMBERSHIP END	CF	AUTHOR	AUTHORSHIP START	AUTHORSHIP END	DETAILS
1	Martin	Makariev	makariev@inrne.bas.bg	INRNE, BAS	BULGARIA	Bulgaria	2018-01-01	-	Yes	Yes	2018-07-01	-	[Edit] [Delete] [Details]
2	Petar	Temnikov	petar.temnikov@gmail.com	INRNE, BAS	BULGARIA	Bulgaria	2018-01-01	-	Yes	Yes	2018-07-01	-	[Edit] [Delete] [Details]
3	Uliases	Barres de Almeida	ulisses@cbpf.br	CBPF	CBPF	Brazil	2018-01-01	-	Yes	Yes	2018-07-01	-	[Edit] [Delete] [Details]
4	Rodrigo Félix	de Araujo Cardoso	rfelix@cbpf.br	CBPF	CBPF	Brazil	2018-01-01	-	No	No	-	-	[Edit] [Delete] [Details]
5	Joao	Giesbrecht Paiva	joao.giesbrecht@gmail.com	CBPF	CBPF	Brazil	2022-01-01	-	Yes	Yes	2022-07-01	-	[Edit] [Delete] [Details]
6	Juan	Bernete Medrano	juan.bernete@ciemat.es	CIEMAT	CIEMAT	Spain	2022-02-20	-	Yes	Yes	2022-08-20	-	[Edit] [Delete] [Details]
7	Alejo	Cifuentes	alejo.cifuentes@ciemat.es	CIEMAT	CIEMAT	Spain	2021-12-09	-	Yes	Yes	2022-06-09	-	[Edit] [Delete] [Details]
8	Juan	Cortina	juan.cortina@ciemat.es	CIEMAT	CIEMAT	Spain	2018-01-01	-	Yes	Yes	2018-07-01	-	[Edit] [Delete] [Details]
9	Carlos	Delgado	carlos.delgado@ciemat.es	CIEMAT	CIEMAT	Spain	2018-01-01	-	Yes	Yes	2018-07-01	-	[Edit] [Delete] [Details]
10	Carlos	Diaz	carlos.diaz@ciemat.es	CIEMAT	CIEMAT	Spain	2018-01-01	-	Yes	Yes	2018-07-01	-	[Edit] [Delete] [Details]

(b) User view.

Figure 3.3 – Comparison of Admin and User views of the Members Page.

Search...

Author: All CF: All

Country: All Group: All Institute: All

Figure 3.4 – Filters for the member list.

- **Group Filter:** This filter allows users to select a specific group. Only members belonging to the selected group will be displayed. By default, all groups are selected.
- **Institute Filter:** This filter narrows down the list of members by selecting a specific institute. It will only show members associated with the selected institute.
- **Authors:** This filter enables users to display only members who are currently authors. The calculation for is an author is done
- **CF:** This filter enables users to display only members who are currently contributing to the Common Fund (CF).

Each of these filters can be selected independently and the Members List will update dynamically as each filter is applied.

The filters for Country, Group, and Institute are interconnected. When a country is selected, the Group filter will display only the groups within that country, and upon selecting a group, the Institute filter will display only institutes associated with the selected group.

To reset filter selections, click on the **Clear Filter** button  located on the right side of the page.

3.3.1 Exporting Data

There is also the option to download the filtered data in several formats, by clicking on the corresponding button    . The export options available are:

- **Copy** (to Clipboard)
- **CSV** Format
- **Excel** Format
- **Print**

These options will export the data currently displayed on the Members List. If filters are applied, only the filtered results will be exported.

3.3.2 Info Button

In the **Details** column of the table, there is an **Info** button  that provides more detailed information about each member. Clicking this button opens a pop-up window (Fig. 3.5) that displays a comprehensive overview of the member's information, organized into the following sections:

- **General:** Displays the member's first name, second name, email address, and role.
- **Current Affiliation:** Shows the member's current institute, group, and country. For former members, a message indicates "No current affiliation."
- **Membership:** Lists all membership periods, including the start and end dates, as well as the institute associated with each period. If no end date is set, "Active" is displayed for the end date.
- **Common Fund:** Indicates whether the member is contributing to the Common Fund and, if applicable, the start date of their contribution.
- **Authorship:** Details the member's authorship periods, including start and end dates for active authorship. If the member has no active authorship but a future period is defined, the start date of the future period is specified. If the member is not an author or no authorship periods are available, a message indicates "No authorship period".
- **Duties:** Details the member's duty periods, including start and end dates. If the member has no duty assigned a message indicates "No duties have been assigned to this member".

MEMBER INFORMATION

✕

GENERAL

First Name

Second Name

Alice

Donini

Email

Role

alice.donini@inaf.it

Researcher

CURRENT AFFILIATION

Institute

Group

Country

OA Roma

INAF

Italy

MEMBERSHIP

Start Date	End Date	Institute
2018-01-01	Active	OA Roma

COMMON FUND

This member is contributing to CF since 2018-01-01

AUTHORSHIP

Start Date	End Date
2018-07-01	Active

ACTIVE DUTIES

Name	Start Date	End Date
Project Managers	2024-05-01	Active
Technical Coordinators	2023-01-01	Active
Common Elements (Data Broker Visualisation)	2023-01-01	Active

Figure 3.5 – Modal showing member information.

3.3.3 Add Member Button

For users with admin permission level, there is the option to add members by clicking the **Add Member** button  on the top left of the page (see Fig. 3.3a). After clicking this button, a modal will open, allowing the user to input the necessary information for adding a new member (see Fig. 3.6a). The requested fields are:

- **First Name** and **Surname** (mandatory)
- **Primary Email** (mandatory)
- **Role**
- **Membership Start** and **Membership End** (the end date can be left blank)
- **Institute** (mandatory)

Additionally, there is a check-box for **Common Found**. If checked, allows setting the CF contribution start date.

3.3.4 Edit Member Button

For users with admin permission level, an **Edit** button  is available in the **Details** column of the table. This button allows users to edit any information related to the member.

In this window you can also assign a duty to a member or end an existing one.

3.4 Authors Page

The **Authors Page**, displayed in Fig. 3.8, is designed to display and manage active authors within the system. This page ensures that all author information, including affiliations and publication-related details, is accurate and consistent for academic and professional use.

+ NEW MEMBER

First Name Surname

Primary Email Role

Membership Start Institute

Membership End Common Found

+ ADD DUTY

SAVE

(a) Modal to add a new member.

EDIT MEMBER

First Name Surname

Primary Email Role

Membership Start Institute

Membership End Common Found

CF Start

CF End

Future Authorship Periods

No future authorship periods logged.

Duties

Duties	Start	End
Project Managers	Start: 2024-05-01	END
Technical Coordinators	Start: 2023-01-01	END
Common Elements (Data Broker visualisation)	Start: 2023-01-01	END
Executive Board	Start: 2021-02-01	END
Telescope Managers	Start: 2021-02-01	END

(b) Modal to edit member's data.

Figure 3.6 – The two modal accessible to manage member information for admin users.

LAMA AUGUST 07, 2025

HOME MEMBERS **AUTHORS** INSTITUTES DUTIES STATISTICS ADMIN PANEL ALICE DONINI LOGOUT

Select Journal: CTAO-113 **EXPORT LIST**

Q: Search: Country: All Group: All Institute: All

#	FIRST NAME	LAST NAME	INSTITUTE	GROUP	COUNTRY	AUTHORSHIP START	AUTHORSHIP END	MAIN AFFILIATION	ACTIONS
1	Abelardo	Moralajo	IFAE-BIST	IFAE	Spain	2018-07-01	-	IFAE-BIST	
2	Abhishek	Abhishek	INFN Siena	INFN	Italy	2023-09-01	-	INFN Siena	
3	Abhishek	Roy	U Hiroshima	JAPAN	Japan	2024-08-01	-	U Hiroshima	
4	Aion	Viana	IFSC-USP	IFSC-USP	Brazil	2025-06-16	-	IFSC-USP	
5	Akira	Okumura	U Nagoya, ISEE	JAPAN	Japan	2018-07-01	-	U Nagoya, ISEE	
6	Alan	Sunny	IAPS Roma	INFN	Italy	2024-07-01	2026-12-31	OA Roma	
7	Alde	Manselli	INFN Roma Tor Vergata	INFN	Italy	2018-07-01	2026-12-31	INFN Roma Tor Vergata	
8	Alessandra	Braccioli	CPPM	CPPM	France	2025-04-01	2026-12-31	CPPM	
9	Alessandra	Lamatra	OA Roma	INFN	Italy	2021-09-01	-	OA Roma	
10	Alessandro	Caroli	OA Roma	INFN	Italy	2018-07-01	-	OA Roma	

Showing 1 to 10 of 280 entries

100 entries per page

Figure 3.7 – Authors page overview.

3.4.1 Author list

The Authors Page provides tools for generating author lists tailored for publication. At the top left of the page, a dropdown menu allows users to select from eight predefined journals (e.g., *Nature*, *Science*, *ApJ*, etc.). After selecting a journal, users can export a .tex file containing the author list formatted according to the chosen journal's guidelines.

To further refine the exported list, a date selector is available, enabling users to specify a cutoff date. Only members who are active authors as of the selected date will appear in the exported list. This functionality ensures that the author list remains accurate and adheres to journal-specific formatting requirements (see Fig. 3.8).

Select Journal

20/01/2025

EXPORT LIST

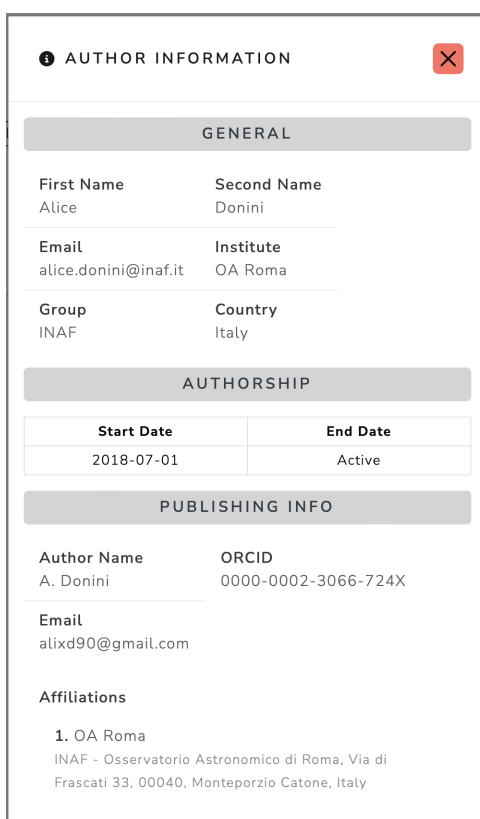
Figure 3.8 – Export options for generating journal-specific author lists.

3.4.2 Info Button

Like for the other pages, The **Info** button , located in the **Details** column of the Authors Page table, provides more detailed information about each author.

Clicking this button opens a pop-up window (Fig. 3.9) displaying detailed data organized into three sections:

- **General:** Displays the author's basic information, including their first and last name, email, group, institute, and country.
- **Authorship:** Lists active and past authorship periods with their start and end dates. If there are no periods, a message indicates 'No authorship period'.
- **Publishing Info:** Provides details for publications, like the author's name, email, ORCID, and affiliations. If multiple affiliations exist, they are displayed in order of appearance, with additional descriptions where available.



The pop-up window displays the following information:

GENERAL	
First Name	Second Name
Alice	Donini
Email	Institute
alice.donini@inaf.it	OA Roma
Group	Country
INAF	Italy

AUTHORSHIP	
Start Date	End Date
2018-07-01	Active

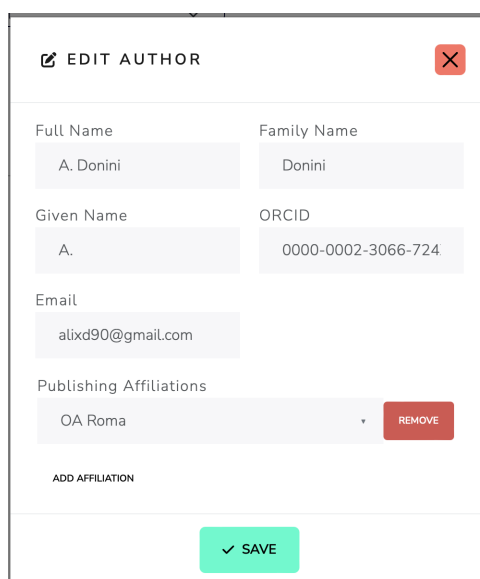
PUBLISHING INFO	
Author Name	ORCID
A. Donini	0000-0002-3066-724X
Email	
alixd90@gmail.com	
Affiliations	
1. OA Roma	
INAF - Osservatorio Astronomico di Roma, Via di Frascati 33, 00040, Monteporzio Catone, Italy	

Figure 3.9 – Pop-up window displaying detailed author information.

3.4.3 Edit Author Button

For users with admin or sapo permission level, an **Edit** button  is available in the **Details** column of the table.

Clicking this button opens a pop-up window (Fig. 3.10) where is possible to edit publishing information related to the author.



The image shows a modal window titled "EDIT AUTHOR" with a close button (X) in the top right corner. The form contains the following fields:

- Full Name:** A. Donini
- Family Name:** Donini
- Given Name:** A.
- ORCID:** 0000-0002-3066-724
- Email:** alixd90@gmail.com
- Publishing Affiliations:** OA Roma (with a dropdown arrow and a red "REMOVE" button)
- ADD AFFILIATION:** A button to add a new affiliation.
- SAVE:** A green button with a checkmark and the text "SAVE" at the bottom.

Figure 3.10 – Pop-up window to edit author information.

3.4.4 Alerts notification

Warnings are displayed at the top of the Authors Page to highlight missing critical information for the author list. These notifications appear as alert boxes and inform administrators if any author is missing essential details, such as their name used in publications or affiliation information. These alerts ensure data gaps are promptly identified and addressed, maintaining the accuracy and completeness of author records (see Fig. 3.11).

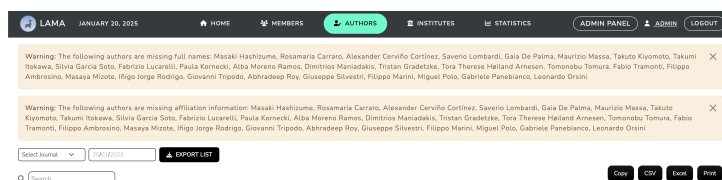


Figure 3.11 – Alerts on the Authors Page highlighting missing details for some authors.

3.5 Institutes Page

The **Institutes Page**, shown in Fig. 3.12, serves as a centralized interface for managing all institutes within the collaboration. Institutes are classified into two categories: official and non-official. Non-official institutes are primarily created for publication purposes and serve as affiliation options for authors.

3.5.1 Add Institute Button

Users with **admin** or **sapo** permissions can add new institutes by clicking the **Add Institute** button **+ New Institute** located at the top left of the page (see Fig. 3.12). Clicking this button opens a modal where users can input the necessary details to create a new institute, as shown in Fig. 3.13.

#	INSTITUTE NAME	LONG NAME	GROUP	COUNTRY	OFFICIAL	LONG DESCRIPTION	ACTIONS
1	ASU	ASU	CZECHIA	Czechia	Yes	Astronomical Institute of the Czech Academy of Sciences, Boct II 1401 - 14100 Prague, Czech Republic	
2	CBPF	CBPF	CBPF	Brazil	Yes	Centro Brasileiro de Pesquisas Físicas, Rua Xavier Sigaud 150, RJ 22290-180, Rio de Janeiro, Brazil	
3	CIEMAT	CIEMAT	CIEMAT	Spain	Yes	CIEMAT, Avda. Complutense 40, 28040 Madrid, Spain	
4	CIEMAT-PIC	CIEMAT-PIC	CIEMAT	Spain	Yes	Port d'Informació Científica, Edifici D, Carrer de l'Albareda, 98193 Bellaterra (Cerdanyola del Vallès), Spain	
5	CPPM	CPPM, CNRS-IN2P3, Aix-Marseille Université	CPPM	France	Yes	Aix Marseille Univ, CNRS/IN2P3, CPPM, Marseille, France	
6	CTAO	Cherenkov Telescope Array Observatory	CTAO	Italy	Yes		
7	DESY	DESY	DESY	Germany	Yes		
8	EPFL	École Polytechnique Fédérale	EPFL	Switzerland	Yes	Laboratory for High Energy Physics, École Polytechnique Fédérale, CH-1015 Lausanne, Switzerland	
9	FESB	FESB	CROATIA	Croatia	Yes	University of Split, FESB, R. Bolševica 32, 21000 Split, Croatia	
10	FZU	FZU	CZECHIA	Czechia	Yes	FZU - Institute of Physics of the Czech Academy of Sciences, Na Slovance 199/2, 182 21 Praha 8, Czech Republic	

Figure 3.12 – Overview of the Institutes Page..

The required fields for adding an institute include:

- **Short Name** (mandatory): serves as a concise identifier for the institute, often used in tables or drop-down selections where space is limited.
- **Full Institute Name** (mandatory): provides the official, complete designation of the institute.
- **Group** (mandatory): This field links the institute to a specific group within the collaboration.
- **Country**: Specifies the country where the institute. If just the group is selected, the country is automatically derived from the group information.
- **Address**: The detailed physical address of the institute. Used in publications.

Additionally, there is a check - box for:

- **Is Official LST**: When selected, the institute is marked as official within the collaboration, and relevant statistics will be available for it.

+ NEW INSTITUTE ✕

Short Name

Full Institute Name

Group

Country

Address

☐ Is official LST

✓ SAVE

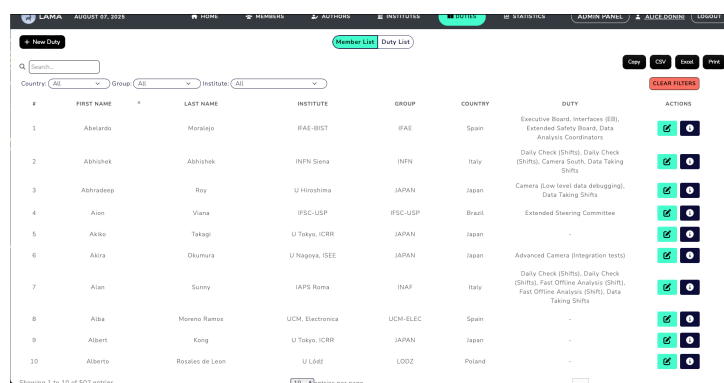
Figure 3.13 – Modal to add a new institute.

3.5.2 Edit Institute Button

For users with either **admin** or **sapo** permissions, an **Edit** button  is available in the **Details** column of the institutes table. Clicking this button opens a modal, allowing users to update any information related to the selected institute (see Fig. ??).

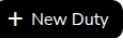
3.6 Duties Page

The **Duties Page** shown in Fig. 3.14 provides an overview on the type of duties available within the collaboration, who is assigned to them, how many positions are available, and other related information.



#	FIRST NAME	LAST NAME	INSTITUTE	GROUP	COUNTRY	DUTY	ACTIONS
1	Abelardo	Morales	IFAE-BIST	IFAE	Spain	Executive Board, Interfaces (IBI), Extended Safety Board, Data Analysis Coordinators	  
2	Abhishek	Abhishek	INFN Siena	INFN	Italy	Daily Check (Shifts), Daily Check (Shifts), Camera South, Data Taking Shifts	  
3	Abhishek	Ray	U Hiroshima	JAPAN	Japan	Camera (Low level data debugging), Data Taking Shifts	  
4	Alan	Viana	IFSC-USP	IFSC-USP	Brazil	Extended Steering Committee	  
5	Akira	Takagi	U Tokyo, ICRR	JAPAN	Japan	-	  
6	Akira	Okumura	U Nagoya, ISEE	JAPAN	Japan	Advanced Camera (Integration tests)	  
7	Alan	Sunny	INFN Roma	INFN	Italy	Daily Check (Shifts), Daily Check (Shifts), Fast Offline Analysis (Shifts), Fast Offline Analysis (Shifts), Data Taking Shifts	  
8	Alba	Moreno Ramos	UCM, Electronica	UCM-ELEC	Spain	-	  
9	Albert	Kong	U Tokyo, ICRR	JAPAN	Japan	-	  
10	Alberto	Rosales de Leon	U Lodz	LODZ	Poland	-	  

Figure 3.14 – General view of the Duties Page.

If the user has admin permissions, they can create new duties using the button located at the top left of the page . The page includes two tabs: **Member List** (the default view) and **Duty List**.

- The **Member List** tab displays a table of all members who have at least one duty assigned. Each entry includes an info button that reveals additional details, such as the member's affiliation and the start date of specific duties. Admin users can edit this information for each entry.
- The **Duty List** tab shows a comprehensive list of all defined duties, along with their descriptions, types, the total number of available positions, and how many members are currently assigned. Fig. 3.15

As with other tables in the application, it is possible to export the data in various formats.

An admin user has the permissions to create new duties Fig. 3.16a, and edit Fig. 3.16c the actual ones, the rest of the users only have permission to view Fig. 3.16b.

3.6.1 Duty Logic Summary

Duties are directly linked to authorship eligibility. A member must have at least one valid duty assignment to appear on the author list.

Duties are categorized into two types: **temporary** and **permanent**.

- **Temporary duties** are automated and valid for two years. Regardless of the exact start date within a year, a short duty is considered valid from **January 1st** of that year

#	DUTY NAME	DESCRIPTION	MAXIMUM SLOTS	ASSIGNED MEMBERS	ACTIONS
31	Container on CAT and Mica Testroom	-	2	0	
32	Coordination	-	1	0	
34	Correction for atmospheric transmission (LST-HRAGC)	Correction for atmospheric transmission using python-LSTAGC-like approach (hence requiring storage space)	1	0	
34	CSS cable monitoring	-	-	0	
35	CSS & CSO cables	-	2	0	
36	CSS locking	-	3	0	
37	DAQ SW development	-	2	0	
38	Data Analysis Coordinators	-	2	0	
39	Data broker SW	-	1	0	
40	Data broker visualisation SW	-	1	1	

Figure 3.15 – View of the Duties List section.

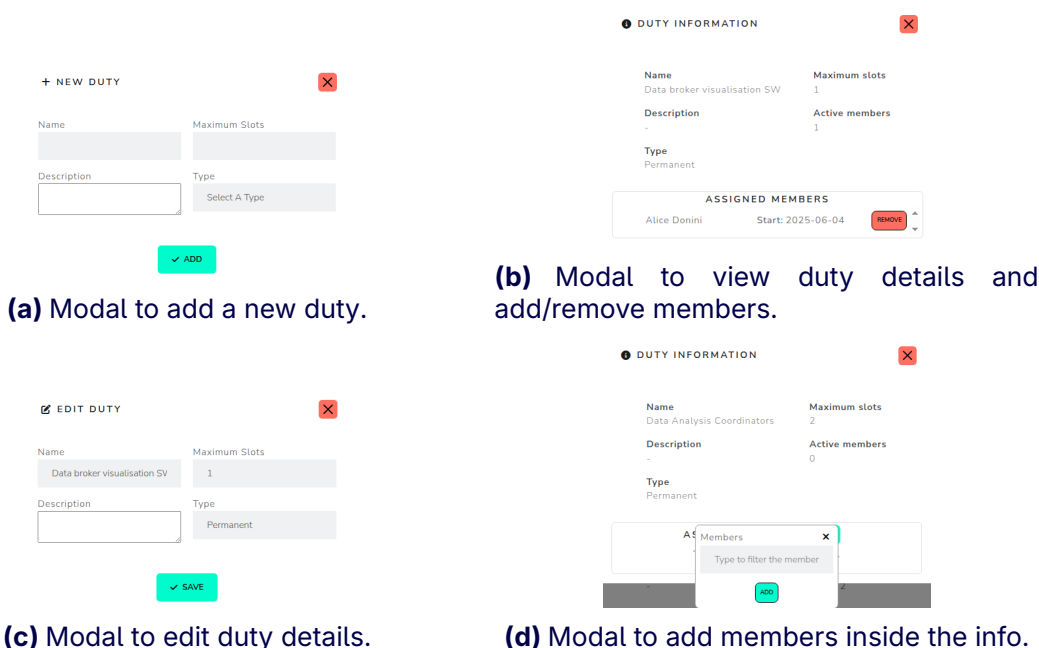


Figure 3.16 – Modals for managing duties, accessible to users with admin permissions.

until **December 31st** of the following year (24 months validity). Examples include shifts, Burst Advocate roles, FOAs, and similar.

- **Permanent duties** begin on a user-defined start date and must be **manually terminated** by specifying an end date. These are typically longer-term responsibilities.

Detailed rules explaining the interaction between duties, Common Fund contributions, membership periods, and authorship periods are presented in Sections 3.9, 3.10, and 3.11.

3.7 Statistics Page

The **Statistics Page** provides visual insights into the collaboration's current status and historical trends. It includes the following components:

- **Summary Cards:** Show the total number of active members, active CF contributors, and active authors.

- **Tables for Countries and Groups:** Summarize the total number of members and authors by country and group, in both absolute numbers and percentages, including their 12-month averages.
- **Trend Analysis:** Interactive charts to visualize membership and CF contribution trends over time.

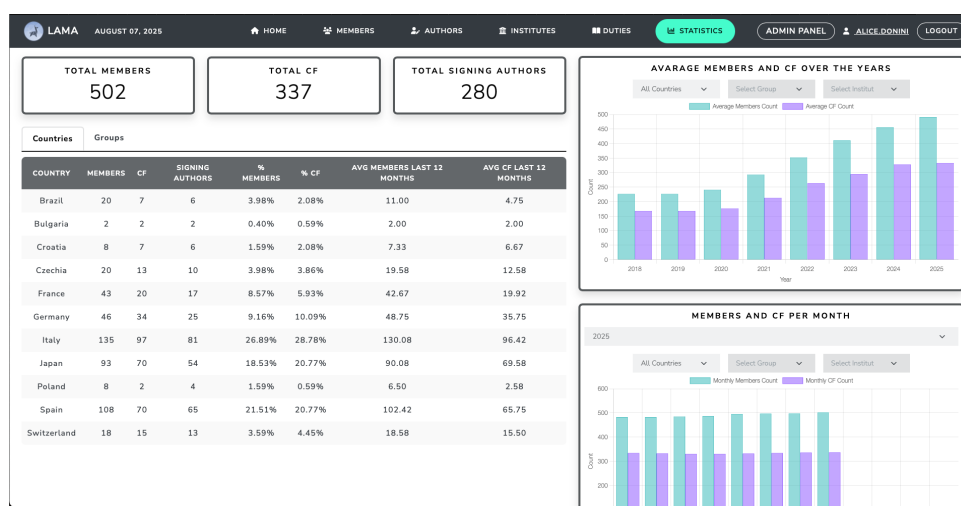


Figure 3.17 – Statistic page view.

3.7.1 Cards

Three cards display the total number of **Members**, **CF** and **Authors**. The totals are calculated dynamically according to the following rules:

- **Total Members:** Count all members whose MembershipPeriod is currently active:
 - start_date is on or before today, and
 - end_date is either not set (null) or later than today.
- **Total CF Contributors:** Count all members whose CommonFund contribution period is currently active and who also have an active or recently active membership:
 - start_date is on or before today, and
 - end_date is either not set (null) or later than today.
- **Total Authors:** Count all members whose AuthorshipPeriod is currently active and who have an active or recently active membership (membership end date no earlier than six months ago):
 - start_date of authorship is on or before today, and
 - end_date is either not set (null) or later than today.

This logic ensures that:

1. Only active records are included in the counts.

2. Members who left less than six months ago are still counted in the “valid” set for authorship and CF statistics, reflecting transitional periods.
3. Data is always up-to-date without manual recalculation.

3.7.2 Tables for Countries and Groups

The data tables for countries and groups show detailed statistics on members and authors over the last 12 months, broken down by the following:

3.7.2.1 1. Countries

The table for countries shows the total number of members, CF contributors and authors per country, their percentage share of the total collaboration, and their average counts over the last 12 months.

- **Members by Country:**
Counts active members based on membership start and end dates relative to the current date. The query filters members whose membership start date is on or before today and whose end date is either null (ongoing) or after today. This ensures only currently active members are counted.
- **CF by Country:**
Counts active Common Fund contributors similarly by their contribution start and end dates relative to today.
- **Authors by Country:**
Counts active authors using authorship start and end dates relative to today. Authors are filtered with start dates on or before today and end dates either null or after today.
- **Percentages:**
For each country, member, CF contributor, and author counts are expressed as a percentage of the total number of active members, CF contributors, and authors in the entire collaboration.
- **Average over 12 months:**
Calculated by counting active members and CF contributors on the 15th of each month over the last 12 months:
 - If today's date is before the 15th, the calculation excludes the current month and starts from the 15th of the previous month.
 - If today's date is on or after the 15th, the calculation includes the current month starting from the 15th.
 - For each monthly date, active members and CF contributors are those whose start date is on or before the 15th and end date is either null or after the 15th.
 - Monthly counts are averaged over the 12 months.

3.7.2.2 2. Groups

The groups table provides a similar breakdown showing total counts, percentages, and 12-month averages of members, CF contributors, and authors per group:

- **Members by Group:**
Counts active members filtered by their membership start and end dates relative to today and grouped by their institute's group.
- **CF by Group:**
Counts active Common Fund contributors similarly grouped by institute group.
- **Authors by Group:**
Counts active authors filtered similarly by authorship start and end dates.
- **Percentages:**
Member, CF contributor, and author counts per group are expressed as a percentage of the collaboration-wide totals.
- **Average over 12 months:**
Calculated identically to the country averages, using the 15th of each month as snapshot dates.

3.7.3 Histograms

There are two histograms on the statistics page: one displays the **Average Members and CF Over the Years**, and the other shows the **Members and CF Per Month** for a selected year.

3.7.3.1 Average Members and CF per Year

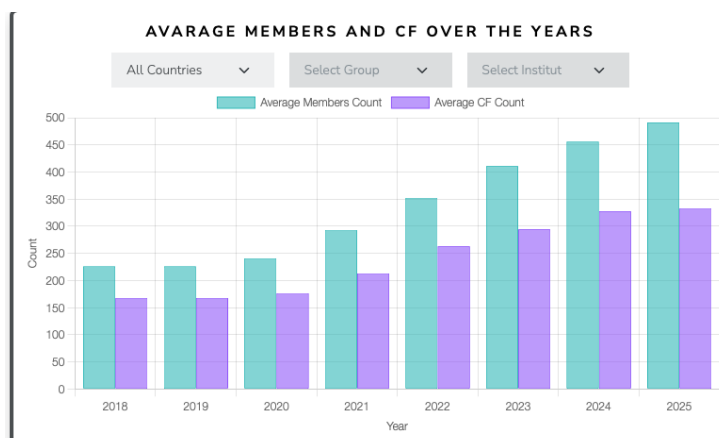


Figure 3.18 – Average Members and Authors Over the Years histogram.

The histogram that displays the average number of members and CF contributors over the years is generated by calculating yearly averages based on the selected filters (such as country, group, and institute). The average number per year is computed by evaluating the active counts on the 15th of each month within that year. The calculation considers whether the year is the current year or a previous one:

- **For the current year:**
The average is based on counts up to the current month. For each month, the active members or CF satisfy:
 - Membership or authorship start date on or before the 15th of the month.

- Membership or authorship end date either null or after the 15th of the month.

Counts are averaged by the number of elapsed months in the current year.

- **For previous years:**

The average is calculated over all 12 months applying the same filtering criteria. The divisor is fixed at 12 months.

The histogram's x-axis shows years (e.g., 2018 to current year) and the y-axis shows average counts. Separate lines represent members and CF to visualize trends.

3.7.3.2 Members and CF Per Month

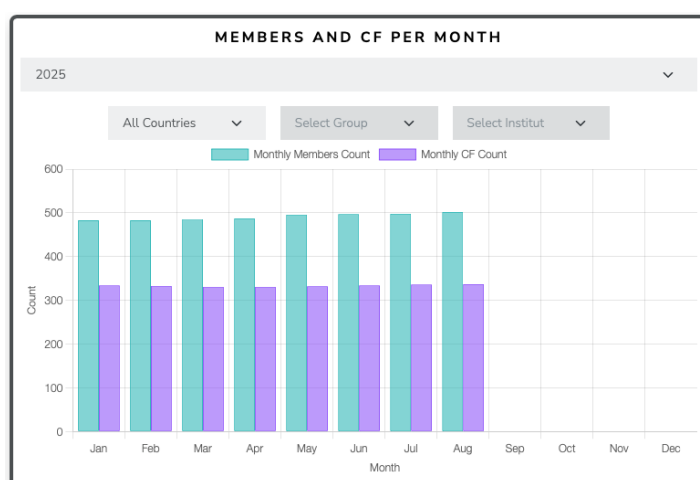


Figure 3.19 – Members and Authors Per Month histogram.

This histogram shows monthly trends for active members and CF contributors in a selected year. Filters such as **Year**, **Country**, **Group**, and **Institute** allow exploration of specific data subsets. Without filters, it displays all members and authors for the current year.

Data Filters: The data displayed in the histogram dynamically adjusts based on the filters selected:

- **Year:** Select a specific year.
- **Country:** Filter by country.
- **Group:** Filter by group.
- **Institute:** Filter by institute.

Monthly Active Count: For each month, active members and CD meet the criteria:

- **Active Members:**
 - Membership start date on or before the 15th of the month.
 - Membership end date either null or after the 15th.
- **Active Authors:**

- Authorship start date on or before the 15th of the month.
- Authorship end date either null or after the 15th.

These criteria ensure accurate monthly counts of contributing members and CF. For the current year, only months up to the current month are included; previous years include all 12 months.

3.8 Admin Panel

For users with Admin permissions, the **Admin Panel** provides access to the Django admin site. Here, administrators can manage system settings, permissions, and perform tasks such as:

- Viewing and editing user roles and permissions.
- Managing group associations and ensuring correct access rights for each user.
- Monitoring and managing all data within the system.

3.9 Common Fund

For a member to be declared as an author, a key requirement is that their institute must contribute its share to the LST Common Fund (CF). A member may have multiple CF periods throughout their time in the collaboration, for instance, if they change institutes or if their institute pauses or resumes its CF contributions.

- The CF contribution periods are tracked independently using the **CommonFund** model.
- By default, the `end_date` of the CF contribution aligns with the `end_date` of the member's **MembershipPeriod**. However, it may end earlier if the institute decides to stop contributing to the CF for that member.
- If a `cf_end` date is defined, then the member's authorship will end six months after that date.

3.10 Membership Rules

Each member must be affiliated with an institute, and their membership is governed by a **MembershipPeriod**. Therefore, if a member is part of the collaboration under different institutes at different times, they will have multiple associated membership periods—one for each institute affiliation.

There are different scenarios in which a member's **MembershipPeriod** may start, end, or change. The following subsections describe the rules for each case.

3.10.1 Starting Membership

This case refers to a person who has never been registered as an LST member.

- A new member must be assigned to an institute and given a `start_date` of the membership.
- A new **MembershipPeriod** is created for the member.
- If the member is designated as an author, also create a **CommonFund** entry and, if applicable (depending on duties), an **AuthorshipPeriod**.

For the rules governing authorship, see Section 3.11.

3.10.2 Changing Institute

If a member changes their institute:

- Their current **MembershipPeriod** is ended one day before the new membership's `start_date`, and a new **MembershipPeriod** is created for the new institute.
- Their active **AuthorshipPeriod**, if any, is evaluated:
 - If they are no longer declared as an author:
 - * The **CommonFund** and **AuthorshipPeriod** are ended.
 - * `AuthorshipPeriod.end_date` is usually set to `membership_end + 6 months`, possibly earlier based on their duty.
 - If they are declared as an author:
 - * If the member has active or future authorship, no new authorship is created (the authorship period continues from the previous institute).
 - * Otherwise, a new **AuthorshipPeriod** and **CommonFund** may be created.

3.10.3 Ending Membership

When a member leaves the collaboration:

- The end date of the membership must be defined.
- If the member is contributing to the Common Fund, that contribution also stops on the same date the membership ends.
- If the member is or was listed as an author (i.e., had a duty and contributed to the Common Fund), their authorship period will automatically end six months after their membership ends.
- Any future authorship is also terminated 6 months after the membership ends.

3.11 Authorship Rules

The start date of a member's authorship depends on when they started contributing to the Common Fund and the timing and type of their duty assignments (see Sec. 3.6.1 for details on duty types). The general idea is that authorship does not begin immediately upon joining, but rather after fulfilling certain contribution and duty requirements. To provide fair compensation, authorship does not end immediately upon leaving the collaboration

or ceasing contribution to the Common Fund; rather, it remains active for an additional six months beyond that date.

If a member is a student, slightly different rules may apply to shorten the waiting period.

The rules described below, which govern authorship eligibility and timing, were established by the Membership Committee. These guidelines reflect the Committee's decisions on how authorship is granted and extended in relation to contributions to the Common Fund and duty assignments. The original document communicating these rules to the collaboration can be accessed here: [TODO!!!! insert link].

3.11.1 Authorship Eligibility

- The member must be contributing to the **Common Fund**.
- The member must have valid duty assignments in the relevant period.
- If changes in authorship or CF status are detected, existing **AuthorshipPeriods** may be updated or ended.

3.11.2 Starting of Authorship

The start of authorship depends on several factors and may vary based on the type of duty and the start date of Common Fund (CF) contribution.

Reminder: Temporary duty validity lasts for two years starting January 1 of the year when the duty took place until 31 Dec of the following year.

When creating a new authorship period:

- **Duty exists before the start of CF contribution:**
The member has a valid duty assignment that started or took place prior to their CF contributions:
 - If the duty started at least six months before the CF began, the authorship period starts on the CF start date.
 - If the duty started less than six months before the CF, the authorship begins six months after the duty start date.
- **Duty starts after the start of CF contribution:**
No past duty exists, but a duty is scheduled to begin after CF contributions have started. In this case, the authorship period starts six months after the future duty start date.
- **No duty assigned:**
If the member has not been assigned any duty, they are not eligible to become an author, and no authorship period is created, even if they are contributing to CF.
- **PhD student case:**
Students are treated differently. Their six-month waiting period starts with their membership. See Sect. 3.11.4.

3.11.3 Ending of Authorship

Authorship eligibility can end based on a member's duty status, membership status, or contribution to the Common Fund. There are three main cases in which an authorship period ends, thus removing a member from the author list:

1. Member still contributing to the Common Fund, but no longer holds a valid duty

- For **temporary** duties, the authorship period ends on December 31 of the year following the year the duty was performed, as explained in Section 3.6.1. For example, if a member only performed a duty in the previous year (2024) and no new duty in the current year (2025), their authorship ends on December 31 of the current year (2025).
- For **permanent duties**, authorship ends on the specified duty end date.
- If a new duty is assigned before the current authorship end date, the system automatically extends the authorship according to the duty validity rules (see Section 3.6.1).
- If no new duty is assigned before the authorship end, the next assignment will trigger the usual authorship start logic, as described in the previous section 3.11.2.
- **Exception, continuity across calendar years:** If a member had a temporary duty in one year, their authorship ends on December 31 of the following year (e.g., duty in 2024 → authorship until end of 2025). If the member is then assigned a new temporary duty in the third year (e.g., 2026), after their previous authorship period has ended, the system **does not treat this as a break**. Instead of applying the standard six-month waiting period, authorship resumes immediately on January 1 of that third year (e.g., 2026), resulting in a continuous authorship period with no gap. This exception applies only to members returning with a temporary duty in the year immediately following the end of their authorship at the close of the previous year.

2. Member stops being declared as an author

In this case the member is no longer contributing to the Common Fund.

- When CF participation ends, the authorship period is extended by a grace period of six months from the CF end date. This applies regardless of the duty type or the previously planned authorship end date.
- If the member has an active authorship period at the time CF participation ends, that period is extended to the CF end date plus six months (even if it was originally set to finish earlier).
- If the member has no active authorship period at the time CF participation ends, but had one at some point during the current CF participation, a new six-month authorship period is granted starting from the CF end date.
- If the member has no active or past authorship period during the current CF participation, no additional authorship period is granted.

3. Membership ends

This case applies when a member leaves the collaboration. Since ending the membership also ends Common Fund participation, the authorship period is extended for a grace period of six months from the membership end date, regardless of duty type. The same rules described above for the end of Common Fund participation apply in this case.

3.11.4 PhD Student Authorship Rules

Students (PhD or Master) have specific rules regarding authorship eligibility and start dates:

- A default 6 - month waiting period from the start of membership applies before authorship can begin.
- If a duty is assigned before completing 6 months of membership, authorship starts only after the 6 - month membership milestone.
- If a duty is assigned after 6 months, authorship starts immediately.
- Rules for ending authorship and membership are the same as for other members.

3.12 Summary of CF, Duty, and Authorship Combinations

3.12.1 Cases in which member is not contributing to CF -> No CF

Duty Type and Timing	Authorship Rule
Any (none, past, or future)	No authorship period is created. CF is a mandatory condition.

3.12.2 Cases in which member is already contributing to CF -> CF ongoing

Duty Type and Timing	Authorship Rule
Permanent duty starts	• Authorship starts on the permanent duty's start date if the CF started more than 6 months earlier (i.e., before duty start minus 6 months). • Authorship starts 6 months after the CF start date if the CF started less than 6 months before the duty start date.
Permanent duty ends	Authorship ends on the permanent duty's end date.
• Temporary duty in previous year • No new duty this year	Authorship ends on December 31 of the current year. If at any time in the current year a duty is assigned, permanent or temporary, the authorship end will be enlarged accordingly.

Continued on next page

Duty Type and Timing	Authorship Rule
- Temporary duty in year N-2 - New temporary duty in year N	Authorship resumes on January 1 of year N, without a waiting period.
- Temporary duty this year - Permanent duty ends	Authorship ends at the end of the next year
- Temporary duty previous year - Permanent duty ends	Authorship ends at the end of the this year

3.12.3 Cases in which a member is declared an author -> CF starts

Duty Type and Timing	Authorship Rule
No duty	No authorship period. Duty is a mandatory condition.
Temporary duty older than N-1 (previous year)	No authorship period is created. Duty is considered too old and is not valid.
Permanent duty	- If the duty started at least 6 months before the CF start: authorship begins on the CF start date. - If the duty started less than 6 months before the CF start: authorship begins 6 months after the duty start date.
Temporary duty previous year (not important if any in the current year)	Authorship starts on CF start date.
Temporary duty current year	- If the CF start date is after July 1 (i.e., more than 6 months after the duty validity start in January), authorship starts on the CF start date. - If the CF start date is before July 1, authorship starts on July 1 (after the 6-month waiting period).

3.12.4 Cases in which a member stops being an author -> CF stops

Duty Type and Timing	Authorship Rule
Ongoing or valid duty	Authorship ends 6 months after the CF end date.
Past duty done after CF start	Authorship ends 6 months after the CF end date. If there was no authorship a new one is created.

3.12.5 Cases in which a member stops membership -> Any CF status

Duty Type and Timing	Authorship Rule
Any duty	Authorship ends 6 months after the membership end date. If there was no active authorship, a new authorship valid for 6 months is created.

3.12.6 Cases for PhD or Master Students

Condition	Authorship Rule
Duty assigned less than 6 months after membership start	Authorship starts after 6 months of membership.
Duty assigned more than 6 months after membership start	Authorship starts immediately.
Membership ends or CF ends	Same rules as in the tables above (e.g., authorship ends 6 months after the membership end date).